Bunyan Yorganci

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EDUCATION

Carleton University, Ottawa, ON — Bachelor of Engineering, Computer Systems Engineering

Sep 2022 – April 2028

Relevant coursework: Operating Systems · Real-Time Concurrent Systems · Intro to Real-Time Systems · Computer Systems Design · Computer Systems Design Lab · Computer Systems Development Project · Computer Organization and Architecture · Algorithms and Data Structures · Object-Oriented Programming · Foundations of Imperative Programming · Intro to Digital Systems · Intro to Software Engineering · Software Validation · Computer Communications · Communication Theory · Systems and Simulation · Circuits and Signals · Probability and Random Processes · Engineering Economics

TECHNICAL SKILLS

- **Languages:** C, C++, Java, Python, Verilog (HDL), ARM assembly, MATLAB, Bash, HTML/CSS
- **Embedded and hardware:** Arduino Due, Raspberry Pi with Sense HAT, TI MSP432 LaunchPad, Xilinx Zynq SoC (Zybo); GPIO, PWM, interrupts, timers, ADC, UART, I²C, SPI, DMA, AMBA/AXI; motor drivers, soldering, circuit troubleshooting
- **Systems:** multithreading and synchronization (C++ threads, mutexes, condition variables), POSIX IPC (fork, shared memory, semaphores), Linux/Unix, UDP sockets, state machines, real-time scheduling (rate-monotonic, WCET analysis)
- **Testing and design:** JUnit 5, Gradle, JaCoCo code coverage, mutation testing, GraphWalker model-based testing, UML class and sequence diagrams
- **Tools:** Git/GitHub, Vivado, Wireshark, MATLAB/Simulink, Logisim, Flask, Socket.IO, OpenCV, SQLite, Firebase, Docker, VMware
- **Networking:** custom DNS (Pi-hole), VPN (WireGuard), static IP routing, network-attached storage
- **Certification:** AWS Certified Cloud Practitioner

PROJECTS

Autonomous Snowplow Robot — SYSC 4805 Computer Systems Design Lab

Fall 2026 (in progress)

- Building, in a team of four, an autonomous Arduino Due robot that clears simulated snow from a taped arena within 5 minutes while avoiding fixed and moving obstacles under a 30 cm/s speed limit
- Developing the motor-driver direction and stop routines for the four-motor platform; rear encoder integration in progress
- Organizing the motor control interface into forward, reverse, turning and stop operations for Arduino-based testing
- Team design integrates four or more sensor types (IMU, line follower, proximity/distance) into a state-machine control architecture, with staged integration and tested modules on GitHub

Software Validation and Boundary Testing — SYSC 4101 Software Validation Fall 2026

- Designed parameterized JUnit 5 boundary tests reaching 100% instruction and branch coverage of Board.withinBorders()
- Covered valid positions and each out-of-range direction; generated JaCoCo coverage reports through Gradle

Operating-System and Interrupt Simulator — SYSC 4001 Operating Systems 2024–25

- Implemented in C++ on Linux a trace-driven simulation of CPU events, interrupts and system calls, with process-control-block state transitions
- Modelled FORK/EXEC, fixed-partition memory allocation and scheduling, and analysed execution traces and I/O scenarios

Linux Process Synchronization — SYSC 4001 Operating Systems Fall 2025

- Wrote multiprocess C programs using fork(), shared memory and semaphores to coordinate access to shared data
- Investigated contention and synchronization behaviour from test output, including a documented livelock case

Multi-threaded Elevator Control System — SYSC 3303 Real-Time Concurrent Systems

Winter 2025

- Built in C++ an elevator scheduler, elevator-car simulator and floor simulator as three separate processes communicating over UDP, able to run across multiple machines
- Implemented a multi-threaded scheduler handling several cars at once, with configurable floor count, elevator count, door timing and travel time
- Delivered the system in iterations, documented with UML class and sequence diagrams that matched the source code

Raspberry Pi Hardware Control and Monitoring — SYSC 3010 Computer Systems Development Project

Winter 2025

- Programmed traffic-light timing and pedestrian-button behaviour in Python on Raspberry Pi GPIO, using assertions to verify hardware control states
- Built browser-to-Sense-HAT LED controls with Flask-SocketIO, plus separate OpenCV monitoring and SQLite logging labs

Motor and Display Hardware Interfaces — ECOR 1044 Mechatronics 2022

- Controlled DC motor direction with PWM, sequenced a stepper motor through an eight-step coil pattern and positioned a servo on Raspberry Pi
- Drove an LCD over I²C to display live updates

Digital Logic and Seven-Segment Decoder — SYSC 2310 Intro to Digital Systems

- Designed gate-level combinational logic in Logisim and verified every four-bit input against the seven-segment display output

LAB AND COURSEWORK EXPERIENCE

- **SoC design on Xilinx Zynq (SYSC 3320):** built combinational and sequential circuits in Vivado, then wrote bare-metal and mixed C/assembly code on the ARM core for interrupt-driven GPIO, AXI timers and UART
- **Embedded real-time programming on MSP432 (SYSC 3310):** programmed digital and analog I/O, hardware and software interrupts and timers in C, and studied race conditions and mutex synchronization with threads
- **Communications simulation (SYSC 3501):** modelled BPSK/QPSK modulation, pulse shaping and channel/SNR effects
- **Systems simulation (SYSC 3600):** modelled dynamic systems in MATLAB and Simulink
- **Computer networks (SYSC 4602):** analysed Wireshark packet traces to examine protocol layering, Ethernet, 802.11 and IPv4

PERSONAL PROJECTS

Home Media and File Server — Ubuntu Server, Docker

- Set up an Ubuntu server running Docker containers, including Jellyfin for media streaming
- Configured network-attached storage for file sharing to Windows machines, a WireGuard VPN for remote access and a static IP, and installed a 1 TB storage expansion

Network-wide DNS Filtering — Raspberry Pi 3B, Pi-hole

- Deployed Pi-hole as a custom DNS server that blocks and filters chosen domains while forwarding valid queries to upstream providers

EXPERIENCE

Computer Refurbisher — Self-employed, Milton, ON **2017 – Present**

- Diagnose and repair used and damaged laptops, replacing screens, batteries, keyboards, I/O controllers and thermal paste
- Upgrade RAM, HDDs, SSDs and batteries, and repair solder joints and blown fuses at board level
- Clone and repair OS installations without data loss; install Windows and Unix systems, including dual-boot setups, for personal use and resale

VOLUNTEERING AND LEADERSHIP

Community Volunteer — Turkish Community, Milton and Edmonton **2017 – Present**

- Help plan and set up community events and cultural festivals, and support activities at Turkish community centres

Cadet — Canadian Armed Forces Cadet Program, Edmonton **2019 – 2020**

- Completed the public speaking program and helped run a recycling fundraiser